

AI In Healthcare, Homework 4: Mimic NLP

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Extract Laryngomalacia notes from database; store in dictionary:

Laryngomalacia is a constriction in the larynx, common in newborns. It usually resolves within a few months of age. My kid had it, and he made cute wheezing noises with every breath for a few weeks.

```
import sqlite_utils
import time
db = sqlite_utils.Database('../mimic3.db')
# "Laryngomalacia is a constriction in the larynx, common in newborns"
disease_name = "Laryngomalacia"
start = time.time()
results = db.query("SELECT * FROM noteevents WHERE LOWER(TEXT) LIKE '%{}%'.format(disease_name))
disease_notes = list(results)
elapsed = time.time() - start
print(f"Found {len(disease_notes)} entries containing {disease_name} in {elapsed:.2f}s")
```

Results: Found 93 entries containing Laryngomalacia in 14.31s

Choose one note to display for demonstration purposes

Once we've extracted a single note, we can refer to it as `demo_note` through the rest of the notebook.

```
# For demonstration purposes, let's choose a single note to display
import random
random.seed(42)
note_index = random.randint(0, len(disease_notes))
demo_note = disease_notes[note_index]["TEXT"]
print(
    f"Displaying note {note_index} out of {len(disease_notes)} notes: \n{demo_note}"
)
```

Extract Note Entities with Spacy and show results

```
import spacy
from prettytable import PrettyTable
from IPython.display import HTML, display

# spacy Entities
print(f"Spacy Entities for demo note {note_index}:\n")
nlp = spacy.load("en_core_web_sm")
doc = nlp(demo_note)
skipped_numbers = 0

# Create the table
table = PrettyTable()
table.field_names = ["Entity Text", "Char Range", "Label"]

skipped_numbers = 0
for ent in doc.ents:
    # Let's exclude numeric entities
    if ent.text[0].isdigit():
        skipped_numbers += 1
        continue
    table.add_row([
        ent.text, f"{ent.start_char}:{ent.end_char}", ent.label_
    ])

# Print the table
# print(table)
display(HTML(table.get_html_string()))
print(f"\n(Skipped {skipped_numbers} numeric entities)")
```

Spacy Entities for demo note 81:

Entity Text	Char Range	Label
Month/Day	110:119	DATE
Month/Day	144:153	DATE
yesterday	175:184	DATE
malacia HR	213:223	PERSON
lib po	280:286	PERSON
MTV	387:390	ORG
PKU	437:440	ORG
today	446:451	DATE
IMP	475:478	ORG
today	547:552	DATE
Monday	576:582	DATE
VNA	583:586	ORG
the next 2 days	599:614	DATE

(Skipped 14 numeric entities)

Use Spacy to run Word2Vec on the Laryngomalacia notes

(Note essentially random similarity values; Spacy model needs more medical knowledge!)

```
# spacy Word2Vec
from gensim.models import Word2Vec
print(f"Spacy Word2Vec for demo note {note_index}:\n")

nlp = spacy.load("en_core_web_sm")

corpus = build_corpus(nlp, disease_notes)

# print(corpus)
print(f"Built corpus from {len(disease_notes)} notes about {disease_name}")

model1 = Word2Vec(corpus, min_count=1)
sub_word = "Stridor"
sub_explanation = "another name for Laryngomalacia"

print(f"\nSimilar words to '{sub_word}', {sub_explanation}:")

rows = model1.wv.similar_by_word(sub_word)
row_getters = [{"Word": lambda r: r[0], "Similarity": lambda r: f"{r[1]: 0.3f}"}]
show_html_table(rows, row_getters)
```

Built corpus from 93 notes about Laryngomalacia

Similar words to 'Stridor', another name for Laryngomalacia:

Word	Similarity
Diffuse	0.327
2189-11-23	0.321
Gastrointestinal	0.309
4p deletion	0.302
40s to 60s	0.294
nasal cannula	0.286
2182-6-10	0.285
68/31	0.283
ECHO	0.282
Discharge Medications	0.279

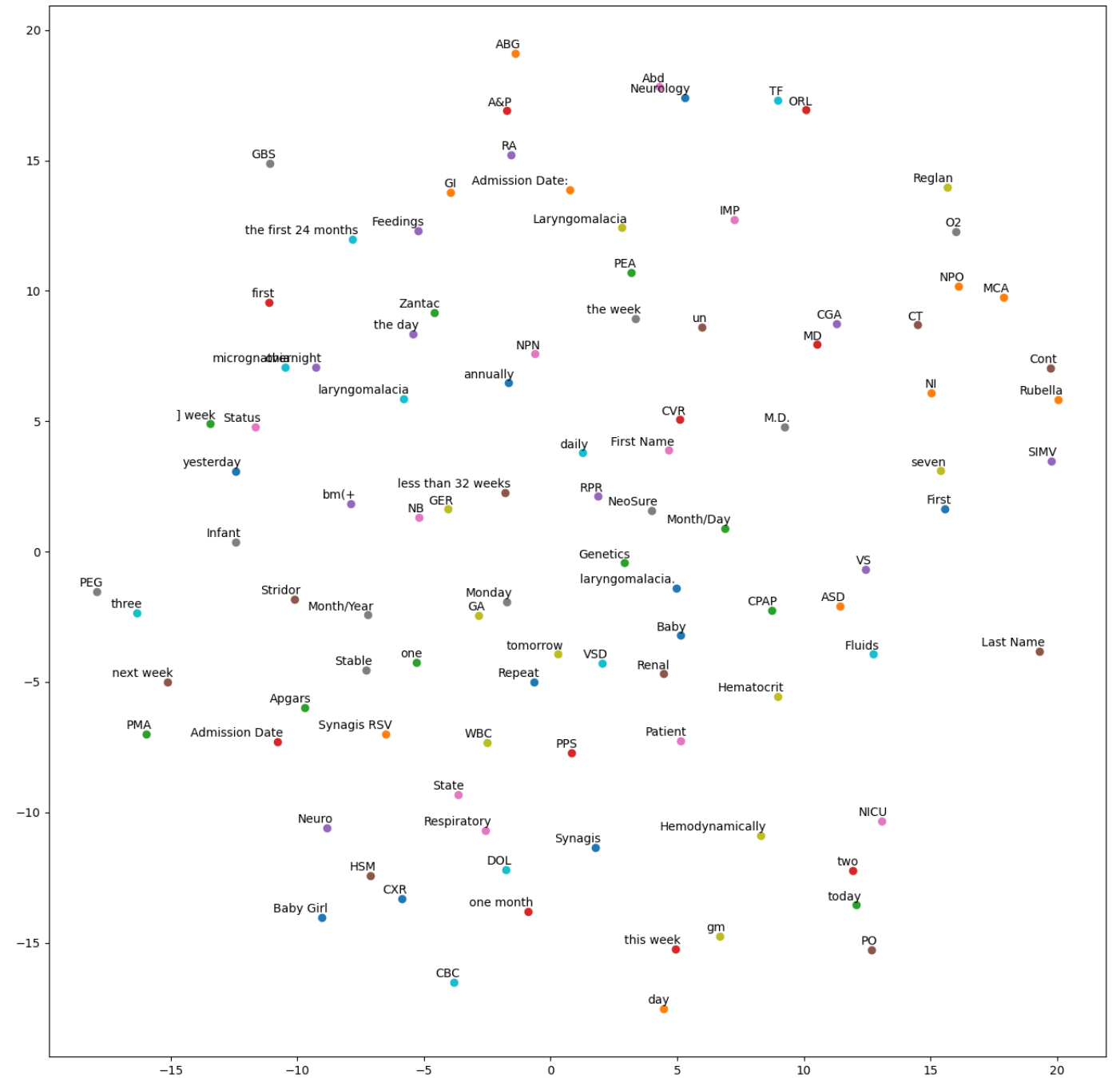
Use Spacy to show tSNE plots of Laryngomalacia notes

```
# spacy tsNE plot
print(f"Spacy tsNE plot for {disease_name} notes:\n")

%matplotlib inline

vocabs = model1.wv.key_to_index.keys()
# remove any words that start with digits
v = [w for w in vocabs if not w[0].isdigit()]
new_v = np.array(list(v)[:100])
tsne_plot(model1, new_v, "spacy_tsne_plot.png")
```

✓ 1.1s



Extract Note Entities with SciSpacy and show results

(Note decreased entity differentiation from default Spacy models)

```
# scispacy Entities
import en_core_sci_lg # A SciSpacy model
nlp = en_core_sci_lg.load()

# spacy Entities
print(f"SciSpacy Entities for demo note {note_index}:\n")
doc = nlp(demo_note)
skipped_numbers = 0

table_rows = [e for e in doc.ents if not e.text[0].isdigit()]
column_fetchers = {
    "Entity Text": lambda e: e.text,
    "Char Range": lambda e: f"{e.start_char}:{e.end_char}",
    "Label": lambda e: e.label_,
}
show_html_table(table_rows, column_fetchers)
```

SciSpacy Entities for demo note 81:

Entity Text	Char Range	Label
Attending	0:9	ENTITY
Note	10:14	ENTITY
Day	15:18	ENTITY
PMA	30:33	ENTITY
room air	51:59	ENTITY
RR	60:62	ENTITY
upper	69:74	ENTITY
airway congestion	75:92	ENTITY
bronch	168:174	ENTITY
yesterday	175:184	ENTITY
laryngeal malacia	203:220	ENTITY
HR	221:223	ENTITY
BP	232:234	ENTITY
ad lib	277:283	ENTITY
night	318:323	ENTITY
E 20	331:335	ENTITY
BM 20 cal/oz	339:351	ENTITY
prilosec	367:375	ENTITY
iron	377:381	ENTITY
MTV	387:390	ENTITY
open crib	406:415	ENTITY
car seat test	423:436	ENTITY
PKU	437:440	ENTITY
hearing screen	459:473	ENTITY
IMP-infant	475:485	ENTITY
laryngomalacia	491:505	ENTITY
stable	513:519	ENTITY
condition	520:529	ENTITY
home today	542:552	ENTITY
Pedi appt	562:571	ENTITY
Monday	576:582	ENTITY
VNA	583:586	ENTITY
days	610:614	ENTITY

Use SciSpacy to run Word2Vec on the Laryngomalacia notes

(Note much more relevant similar words; this medical model knows more!)

```
# Scispacy Word2vec
print(f"SciSpacy Word2Vec for demo note {note_index}:\n")
nlp = spacy.load("en_ner_bc5cdr_md")

corpus = build_corpus(nlp, disease_notes)

# print(corpus)
print(f"Built corpus from {len(disease_notes)} notes about {disease_name}")

model1 = Word2Vec(corpus, min_count=1)
sub_word = "Stridor"
sub_explanation = "another name for Laryngomalacia"

print(f"\nSimilar words to '{sub_word}', {sub_explanation}:")

rows = model1.wv.similar_by_word(sub_word)
row_getters = {"Word": lambda r: r[0], "Similarity": lambda r: f"{r[1]: 0.3f}"}
show_html_table(rows, row_getters)
```

SciSpacy Word2Vec for demo note 81:

Built corpus from 93 notes about Laryngomalacia

Similar words to 'Stridor', another name for Laryngomalacia:

Word	Similarity
ventriculomegaly	0.273
esophogram	0.267
RhoGAM	0.232
dysgenesis of the corpus callosum	0.231
O2 sats	0.220
RA	0.217
cleft of the soft palate	0.215
glaucoma	0.211
hypothermia	0.209
nystatin ointment	0.203

Use SciSpacy to show tSNE plots of Laryngomalacia notes

```
# scispacy tSNE plot
print(f"SciSpacy tSNE plot for demo note {note_index}:\n")

vocabs = model2.wv.key_to_index.keys()
# remove any words that start with digits
v = [w for w in vocabs if not w[0].isdigit()]
new_v = np.array(list(v)[:100])
tsne_plot(model1, new_v, "assets/scispacy_tsne_plot.png")
```

✓ 0.9s

